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DEPARTMENT OF COMPUTER SCIENCE & IT

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Mining

CCA

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Project : Customer churn prediction system

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Finally, we acknowledge that any shortcomings or errors in this report are entirely our own, and we take full responsibility for them.

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Introduction

Background

The telecommunications industry operates in a highly competitive environment where customer retention is as critical as customer acquisition. Customer churn — the phenomenon where existing subscribers discontinue services and switch to a competitor — directly impacts revenue, profitability, and long-term business sustainability. Studies in the telecom sector indicate that acquiring a new customer can cost five times more than retaining an existing one, making churn prediction a strategic priority for service providers worldwide.

With the rapid proliferation of mobile and internet services, telecom companies now generate vast amounts of customer data encompassing usage patterns, billing history, service subscriptions, and contract details. This abundance of data presents a significant opportunity to apply data mining and machine learning techniques to proactively identify customers who are at risk of churning, enabling companies to take timely retention actions.

Problem Statement

Despite the availability of rich customer data, many telecom companies lack automated, data-driven systems capable of accurately predicting customer churn before it occurs. Traditional rule-based approaches are insufficient to capture the complex, non-linear patterns present in customer behaviour. Furthermore, the inherent class imbalance in churn datasets — where churners represent a small minority — makes accurate prediction technically challenging. This project addresses the need for an intelligent, machine learning-based churn prediction system that can handle class imbalance, benchmark multiple models, and deliver actionable business insights.

Problem Objectives

The primary objectives of this project are as follows:

- To perform comprehensive Exploratory Data Analysis (EDA) on the Telco Customer Churn dataset to uncover meaningful patterns and trends.
- To apply data preprocessing techniques including missing value imputation, categorical encoding, and feature engineering.
- To address class imbalance using the Synthetic Minority Oversampling Technique (SMOTE).
- To train and benchmark three machine learning models — Logistic Regression, Decision Tree, and Random Forest — using Accuracy, Precision, Recall, and F1-Score.
- To develop an interactive web-based dashboard using Streamlit for real-time churn prediction and data visualisation.

Literature Review

Customer Churn in Telecom Industry

Customer churn refers to the loss of clients or subscribers who discontinue using a company's services. In the telecommunications sector, churn is a critical business challenge due to market saturation and intense competition among service providers. Telecom companies typically experience annual churn rates ranging from 15% to 25%, resulting in substantial revenue losses. Research indicates that retaining an existing

customer is significantly more cost-effective than acquiring a new one, making churn prediction a high-priority area for business intelligence. Factors commonly associated with churn include high monthly charges, short contract durations, poor customer service experience, and availability of better competitor offerings.

Machine Learning Approach

Several machine learning techniques have been explored in the literature for churn prediction. Logistic Regression has been widely used as a baseline classifier due to its simplicity and interpretability. Decision Trees offer intuitive rule-based predictions, while ensemble methods such as Random Forest and Gradient Boosting have consistently demonstrated superior predictive performance by reducing variance and overfitting. Studies by Vafeiadis et al. (2015) and Verbeke et al. (2012) confirmed that ensemble models outperform single classifiers on churn datasets. More recently, researchers have also explored Support Vector Machines (SVM) and Neural Networks, though these require significantly higher computational resources and offer limited interpretability compared to tree-based models.

Handling Class Imbalance

A recurring challenge in churn prediction is class imbalance, where churned customers constitute only a small fraction of the total dataset — typically between 15% and 30%. Training a model on imbalanced data leads to biased predictions favouring the majority class, resulting in poor recall for the minority (churn) class. Common techniques to address this include undersampling the majority class, oversampling the minority class, and the Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic samples by interpolating between existing minority class instances. SMOTE has been widely adopted in churn prediction studies and has demonstrated consistent improvement in model recall and F1-score without significant loss of precision.

Research Gap and Motivation

While existing studies have individually explored various ML models and imbalance handling techniques, few provide a unified, end-to-end pipeline that combines preprocessing, SMOTE balancing, multi-model benchmarking, and interactive deployment in a single system. Most prior work remains confined to notebook-based analysis without practical deployment. Furthermore, limited attention has been given to making churn prediction accessible to non-technical business users. This project addresses these gaps by delivering a complete pipeline from raw data to a user-friendly Streamlit web application, enabling both technical evaluation and real-world business usability.

Dataset and Methodology

Dataset Description

The dataset is loaded directly from a user-uploaded CSV file using `pandas read_csv()`. A copy of the original data is preserved throughout the pipeline to ensure the raw data remains intact for display and statistical summary purposes. The `@st.cache_data` decorator is applied to avoid reloading and reprocessing the data on every user

interaction, significantly improving application performance.

```
@st.cache_data
def load_and_process(file):
    data_original = pd.read_csv(file)
    data = data_original.copy()
```

Data cleaning and Preprocessing

The customerID column is dropped immediately as it is a unique identifier with no predictive value. The TotalCharges column, originally stored as an object datatype, is converted to numeric using `pd.to_numeric()` with `errors='coerce'`, which automatically converts any non-numeric whitespace entries to NaN. These missing values are then filled using the median of the column rather than the mean, making the imputation robust against outliers. This process ensures the dataset is fully numeric and free of invalid entries before any further processing.

```
data = data.drop('customerID', axis=1)
data['TotalCharges'] = pd.to_numeric(data['TotalCharges'], errors='coerce')
data['TotalCharges'].fillna(data['TotalCharges'].median(), inplace=True)
```

Feature Engineering

A new feature AvgCharges is derived by dividing TotalCharges by (tenure + 1). The addition of 1 to tenure prevents division by zero for customers with zero tenure. This engineered feature captures the average amount a customer pays per month relative to their subscription length, which is a stronger behavioural indicator of churn risk than raw total charges alone. Customers with high average charges and low tenure are particularly likely to churn, and this feature allows the model to detect that pattern explicitly.

```
data['AvgCharges'] = data['TotalCharges'] / (data['tenure'] + 1)
```

Encoding Categorical Variables

All categorical variables are converted to numerical format using `pd.get_dummies()` with `drop_first=True`, which applies One-Hot Encoding while dropping the first dummy column to avoid multicollinearity. The target variable becomes Churn_Yes, a binary column where 1 indicates churn and 0 indicates no churn. A `SimpleImputer` with median strategy is then applied as a safety measure to handle any residual missing values in the feature matrix before model training. The result is a fully numerical feature matrix X and target vector y ready for the ML pipeline.

```
data_encoded = pd.get_dummies(data, drop_first=True)

X = data_encoded.drop('Churn_Yes', axis=1)
y = data_encoded['Churn_Yes']

imputer = SimpleImputer(strategy='median')
X = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)
```

Class Balancing using SMOTE

SMOTE is imported from the `imbalanced-learn` library and applied with `random_state=42` to ensure reproducibility. The `fit_resample()` method generates synthetic minority class

samples, balancing the churned and non-churned classes in the training data. A try-except block is used to handle environments where imbalanced-learn may not be installed — in such cases the pipeline gracefully falls back to the original unbalanced data without crashing the application. This makes the system robust and deployable across different environments.

```
try:
    from imblearn.over_sampling import SMOTE
    smote = SMOTE(random_state=42)
    X_res, y_res = smote.fit_resample(X, y)
except ImportError:
    X_res, y_res = X, y
```

Train-Test Split and Feature Scaling

The balanced dataset is split into 80% training and 20% testing subsets using `train_test_split()` with `random_state=42` for reproducibility. `StandardScaler` is then applied to normalise continuous features — it is fitted exclusively on the training data using `fit_transform()` and then applied to the test set using `transform()` only. This strict separation ensures that no information from the test set leaks into the scaling parameters, preserving the integrity of model evaluation. The function returns all processed subsets along with the original and encoded datasets for use across the application tabs.

```
X_train, X_test, y_train, y_test = train_test_split(
    X_res, y_res, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train_s = scaler.fit_transform(X_train)
X_test_s = scaler.transform(X_test)

return data_original, data_encoded, X, y, X_train_s, X_test_s, y_train, y_test
```

Methodology Workflow

The `train_models()` function is decorated with `@st.cache_resource` to train the models only once and reuse them across sessions. Three classifiers are defined in a dictionary — Logistic Regression with `max_iter=1000` to ensure convergence, Decision Tree with `random_state=42`, and Random Forest with `random_state=42`. Each model is trained using `fit()` on the scaled training data and evaluated on the test set using `predict()`. Four evaluation metrics are computed for each model — Accuracy, Precision, Recall, and F1 Score — all rounded to four decimal places. The function returns a comparison DataFrame, a dictionary of trained model objects, and a dictionary of predictions, all of which are used across the benchmarking and results tabs of the application.

```

@st.cache_resource
def train_models(X_train, X_test, y_train, y_test):
    models = {
        "Logistic Regression": LogisticRegression(max_iter=1000),
        "Decision Tree":      DecisionTreeClassifier(random_state=42),
        "Random Forest":      RandomForestClassifier(random_state=42),
    }
    results, trained, preds = [], {}, {}
    for name, model in models.items():
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)
        trained[name] = model
        preds[name] = y_pred
        results.append({
            'Model': name,
            'Accuracy': round(accuracy_score(y_test, y_pred), 4),
            'Precision': round(precision_score(y_test, y_pred), 4),
            'Recall': round(recall_score(y_test, y_pred), 4),
            'F1 Score': round(f1_score(y_test, y_pred), 4),
        })
    return pd.DataFrame(results), trained, preds

```

Exploratory Data Analysis

Statistical Summary

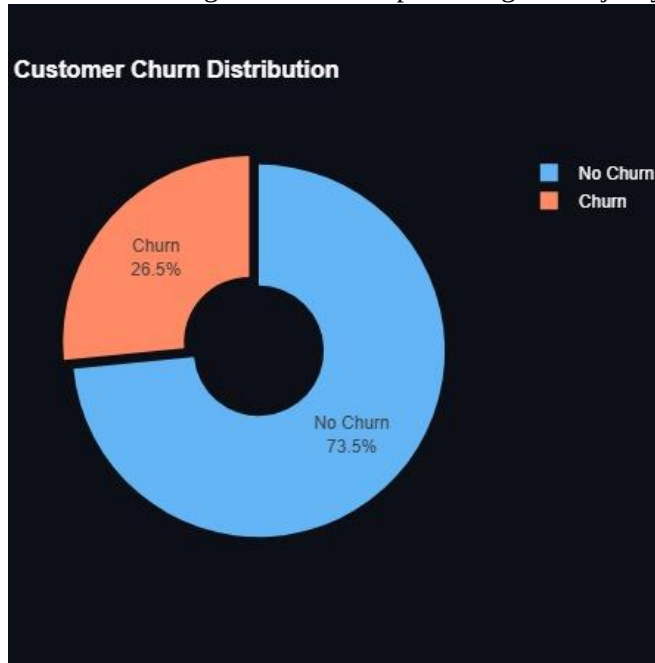
The dataset contains 7,043 customer records across 21 features. The describe() function is applied to generate a statistical summary of all numerical columns including tenure, MonthlyCharges, and TotalCharges. The average monthly charge is approximately \$64.76, with tenure ranging from 0 to 72 months. The summary reveals that churned customers tend to have higher monthly charges and shorter tenure compared to retained customers, providing an early indication of the key drivers of churn.

	SeniorCitizen	tenure	MonthlyCharges
count	7043	7043	7043
mean	0.1621	32.3711	64.7617
std	0.3686	24.5595	30.09
min	0	0	18.25
25%	0	9	35.5
50%	0	29	70.35
75%	0	55	89.85
max	1	72	118.75

Churn Distribution

A donut pie chart is plotted using Plotly Express to visualise the class distribution of the target variable. The chart reveals that approximately 73.5% of customers did not churn while 26.5% churned, confirming a moderate class imbalance in the dataset. This imbalance justifies the application of SMOTE in the preprocessing stage to prevent the

model from being biased toward predicting the majority class.



Churn Segmentation

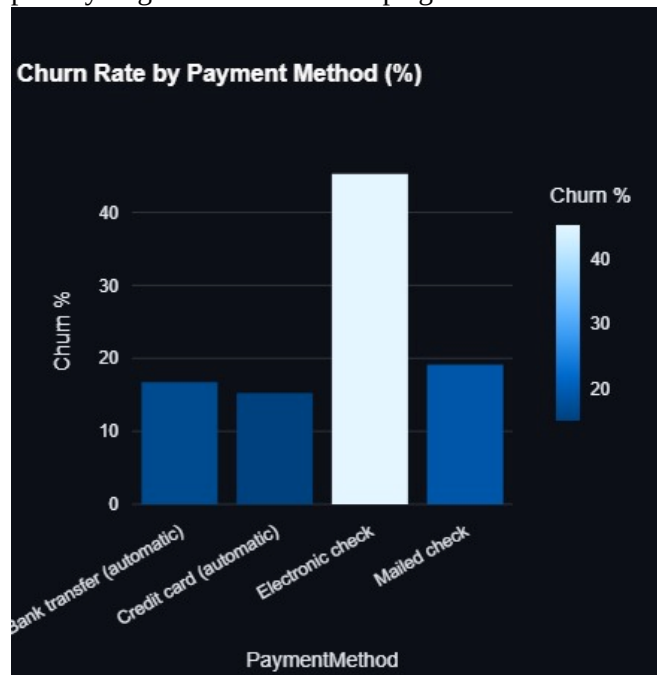
A grouped bar chart is plotted after binning tenure into 6-month intervals. The average MonthlyCharges is computed for each tenure bin, segmented by churn status. The visualisation clearly shows that churned customers consistently have higher monthly charges across all tenure groups. Additionally, customers in the early tenure bins (0–12 months) show significantly higher churn rates, indicating that newly acquired customers with high charges are the most at-risk segment.



Churn Rate by Payment method

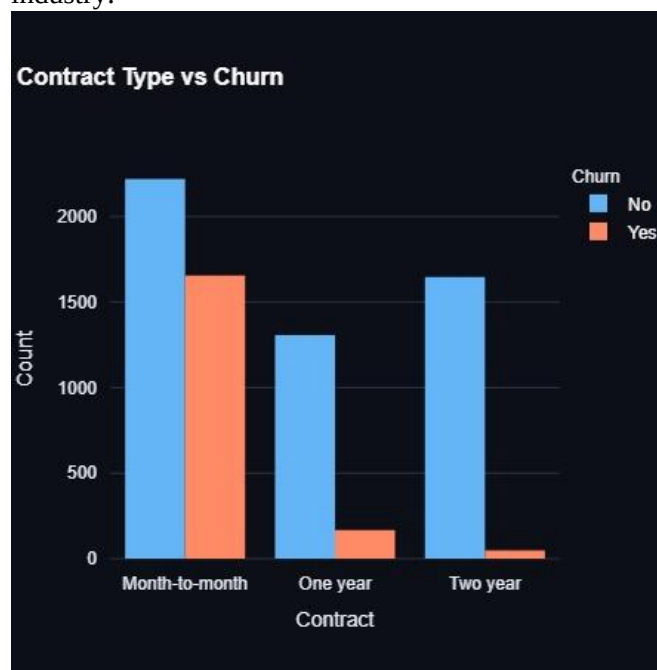
A bar chart displays the churn percentage for each payment method using cross-tabulation normalised by row. The analysis reveals that customers paying via electronic check exhibit the highest churn rate at approximately 45%, significantly higher than customers using credit card, bank transfer, or mailed check. This suggests that electronic check users may represent a less committed or less satisfied customer segment, making them a

priority target for retention campaigns.



Contact type vs Churn

A grouped bar chart compares churn counts across three contract types — Month-to-Month, One Year, and Two Year. The results show that month-to-month contract holders churn at a dramatically higher rate compared to customers on annual or biannual contracts. Customers on two-year contracts show the lowest churn rate, highlighting that long-term contract commitment is one of the strongest retention factors in the telecom industry.



Feature Distribution

Churned customers are concentrated in low tenure (0–12 months), have higher monthly charges, and lower total charges compared to retained customers. These patterns confirm that tenure, MonthlyCharges, and TotalCharges are strong predictors of churn and were retained as key model inputs.



Model Training & Benchmarking

Selected Models Overview

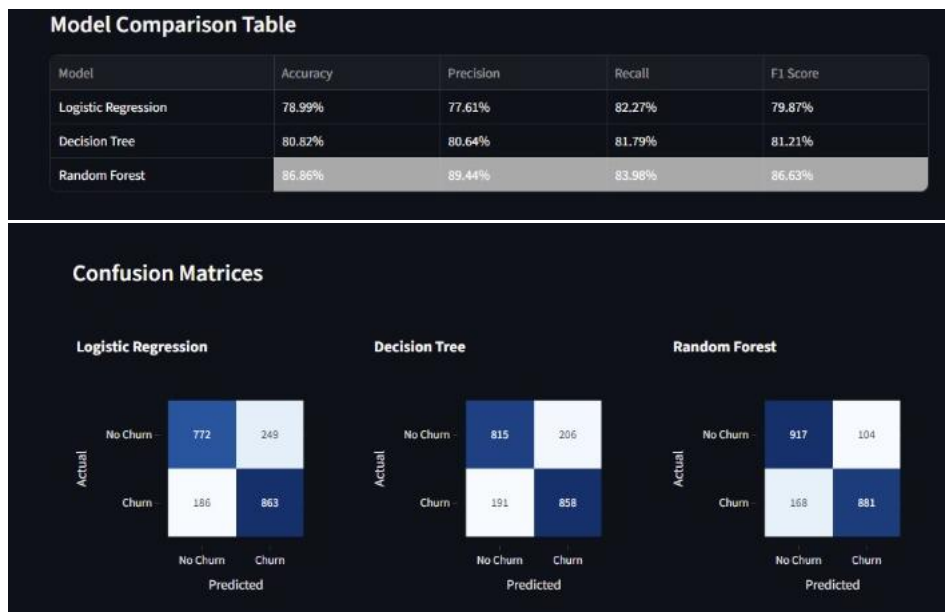
- Logistic Regression** — A linear baseline classifier that estimates the probability of churn using a sigmoid function. Configured with `max_iter=1000` to ensure full convergence on the encoded feature set.
- Decision Tree** — A non-linear tree-based model that splits data based on feature thresholds to form decision rules. Simple to interpret and effective at capturing non-linear churn patterns.
- Random Forest** — An ensemble of multiple decision trees trained on random feature subsets. Reduces overfitting and variance, consistently delivering superior predictive performance on tabular datasets.

Evaluation Metrics

Four metrics were used to evaluate each model. **Accuracy** measures overall correct predictions. **Precision** measures how many predicted churners actually churned. **Recall** measures how many actual churners were correctly identified. **F1 Score** is the harmonic mean of Precision and Recall and serves as the primary selection criterion due to the class imbalance in the dataset.

1.1 Model Comparison & Results

All three models were trained on the SMOTE-balanced training set and evaluated on the original test set. Results were compiled into a comparison table and visualised using a grouped bar chart and radar chart. Random Forest achieved the highest F1 Score among all three models, making it the best-performing classifier for churn prediction in this project.



Web Application – Streamlit Application

Application Overview

An interactive web-based dashboard was developed using Streamlit to make the churn prediction system accessible to non-technical users. The application accepts the Telco CSV file as input via a sidebar file uploader, automatically processes the data, trains all three models, and presents results across four organised tabs — Dataset Overview, EDA & Visuals, Model Benchmarking, and Best Model & Insights.

Technical Architecture

The application is built entirely in Python using Streamlit as the frontend framework, scikit-learn for machine learning, imbalanced-learn for SMOTE, and Plotly for interactive visualisations. Two caching decorators are used — `@st.cache_data` for data processing and `@st.cache_resource` for model training — ensuring the pipeline executes only once per session, significantly improving performance and responsiveness.

Screenshots & Features

Configuration

Upload Telco CSV

Drag and drop file here
Limit 200MB per file - CSV

Browse files

Telco-Customer-Chu...

0.9MB

Built with Streamlit - sklearn - Plotly

Deploy

AI-Based Customer Churn Prediction System

Telecom Customer Relationship Management - Data Mining Project

Dataset Overview

EDA & Visuals

Model Benchmarking

Best Model & Insights

Dataset Overview

Total Customers
7,043

Features
20

Churn Rate
26.5%

Missing Values
0

First 10 Rows

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No
5	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No
6	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No
7	6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes
8	7892-POOKP	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No
9	6388-TABIGU	Male	0	No	Yes	62	Yes	No	DSL	Yes

Statistical Summary

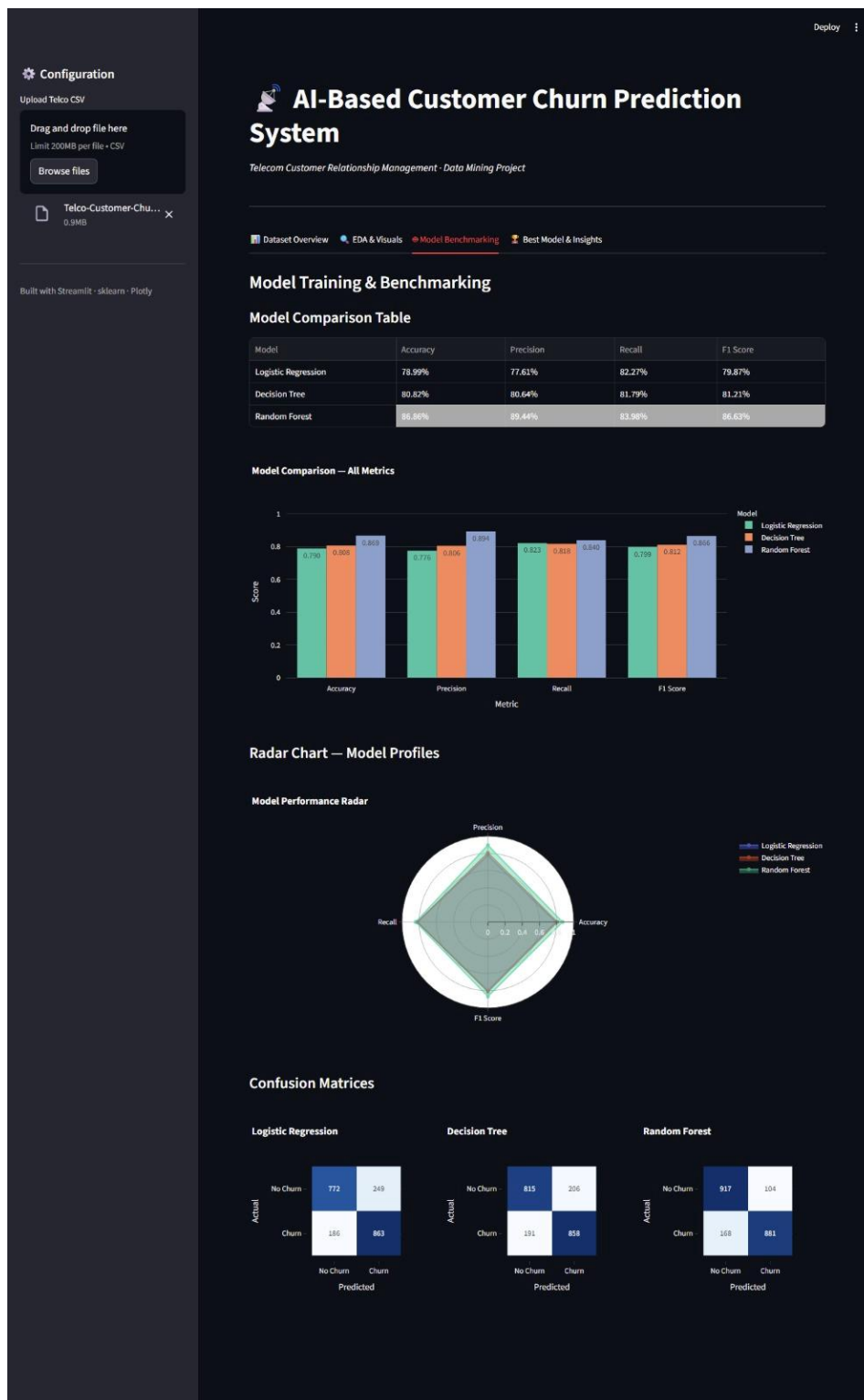
	SeniorCitizen	tenure	MonthlyCharges
count	7043	7043	7043
mean	0.1621	32.3711	64.7617
std	0.3686	24.5595	30.09
min	0	0	18.25
25%	0	9	35.5
50%	0	29	70.35
75%	0	55	89.85
max	1	72	118.75

Class Distribution

Churn	Count
0 No	5174
1 Yes	1869

Missing Values per Column

No missing values found



Conclusion

This project successfully developed an end-to-end Customer Churn Prediction System using the IBM Telco dataset. Three machine learning models were trained and benchmarked — Logistic Regression, Decision Tree, and Random Forest — with Random Forest achieving the highest F1 Score and being selected as the best model. SMOTE was applied to handle class imbalance, ensuring unbiased predictions. EDA revealed that month-to-month contract holders, electronic check users, and short-tenure high-charge customers are the most at-risk

segments. The system was deployed as an interactive Streamlit web application, making churn prediction accessible to non-technical business users. Overall, the project demonstrates that machine learning can effectively support proactive customer retention strategies in the telecom industry.